Arista Hebbal Edge Access Node HEB-ACC-25 QoS Congestion Event Resolution SOP

Node: HEB-ACC-25 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_3754

Description

The Hebbal Edge Access Node HEB-ACC-25, operating under the Arista CLI dialect, is prone to Quality of Service (QoS) congestion events stemming from microbursts, interface tail drops, and QoS queue starvation. This phenomenon arises due to the inherent characteristics of the Access role on Arista equipment, where the emphasis on providing maximum flexibility and scalability in network design can sometimes lead to inefficient queuing mechanisms and inadequate buffer management. The occurrence of microbursts and interface tail drops can be attributed to the interplay between the network's capacity to handle varying traffic loads and the time-sensitive nature of critical network services. Furthermore, QoS queue starvation can result from an imbalance in traffic distribution and policing, where some flows are given preferential treatment over others, leading to an accumulation of packets in specific queues. This can ultimately cause the network to become congested, leading to elevated latency, packet loss, and poor overall network performance. The resolution of QoS congestion events on Arista equipment necessitates a thorough understanding of the underlying network architecture, traffic patterns, and QoS configuration to effectively mitigate these issues and restore optimal network functionality.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and that all necessary personnel are present to facilitate the resolution process.

- Check the status of redundant links to ensure that they are functioning correctly and providing an alternate path for critical network services.

- Confirm that all necessary network access and authentication credentials are in place to facilitate remote access to the Arista equipment.

Potential Root Causes

Cause	Probability
Microbursts and Interface Tail Drops	High
QoS Queue Starvation	Medium
Inadequate Buffer Management	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface status, Expected Output: Interface Status: Up/Down/Unknown, Description: This command is used to verify the operational status of all interfaces on the Arista equipment,
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including those associated with the Hebbal Edge Access Node
HEB-ACC-25. The output will indicate whether the interfaces are up,
down, or unknown, providing valuable information for identifying
potential issues related to microbursts and interface tail drops.

Command: show qos queue stats, Expected Output: Queue Statistics:
Packets Forwarded/Dropped/Buffered, Description: This command is used
to gather information about the QoS queue statistics on the Arista
equipment, including the number of packets forwarded, dropped, and
buffered. This data will aid in identifying potential issues related
to QoS queue starvation and inadequate buffer management.
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Resolution Steps

Step 1: Step: Adjust QoS Queue Configuration, Command: config qos queue , Description: This step involves adjusting the QoS queue configuration to ensure that the queue is properly sized and configured to handle the expected traffic load. The command is used to adjust the bandwidth and burst size of the specified queue, allowing for more efficient queuing and minimizing the occurrence of QoS queue starvation.

Step 2: Step: Implement Traffic Policing, Command: config policer , Description: This step involves implementing traffic policing to regulate the traffic flow and prevent excessive bandwidth utilization. The command is used to create a policer with the specified rate and burst size, allowing for more effective policing and preventing QoS queue starvation.

Post-Check Verification

```
Command: show qos queue stats, Expected Output: Queue Statistics:
Packets Forwarded/Dropped/Buffered, Description: This command is used
to verify the effectiveness of the resolution steps by checking the
QoS queue statistics. The output will indicate whether the number of
packets forwarded, dropped, and buffered has improved, providing
valuable information for confirming that the resolution steps have
been successful.
```

Roll-back Procedures

- Step: Restore Original QoS Queue Configuration, Command: config qos queue , Description: This step involves restoring the original QoS queue configuration to its previous state in case the resolution steps failed to resolve the issue. The command is used to restore the original bandwidth and burst size of the specified queue, ensuring that the network returns to its previous state.

- Step: Remove Implemented Traffic Policing, Command: no config policer , Description: This step involves removing the implemented traffic policing configuration to prevent any potential issues associated with policing. The command is used to delete the policer with the specified name, ensuring that the network returns to its previous state.